AGRICONNECT TN

A direct, price-transparent marketplace connecting Tamil Nadu's farmers to buyers

Project report — problem statement, existing systems, and proposed solution

· Spring Boot backend · Built on free-tier infrastructure

1. Problem statement

Tamil Nadu's farmers routinely sell their harvest without knowing what it is actually worth. The produce moves through several hands — a commission agent, a wholesale buyer, sometimes a second reseller — before it reaches a consumer, and the farmer who grew it typically sees only the first, smallest cut of that chain.

A real example

A mango farmer near Dindigul harvests his crop each summer. He hires a tempo and drives to the mandi with no way to know what price nearby mandis are offering that day, and no buyer arranged in advance. At the mandi, a commission agent takes the produce, deducts a commission (typically 6–10%) plus market fees, and resells it onward — the farmer never learns the final resale price. Because mangoes are perishable and the transport cost is already sunk, the farmer has almost no bargaining power once he arrives.

₹10/kg	Typical price paid to the farmer at the mandi
₹40/kg	Typical price paid by the end consumer for the same produce
6–10%	Commission agent cut, before additional market fees

The core problem is not lack of market access — farmers already travel to mandis every season. It is that they travel blind: no price comparison across markets, no pre-arranged buyer, and no leverage once perishable produce is already in transit.

2. Existing systems and their gaps

Several systems already attempt to solve parts of this problem. Understanding exactly where each one falls short is what shaped this project's design.

System	Who runs it	Key gap
e-NAM	Government of India	Only ~23% of mandis connected; interstate trade volume fell 78% in FY25. Farmers cite immediate cash needs and distrust of delayed digital payment.
Uzhavan app	Tamil Nadu Government	Mainly an information/subsidy app, not a live price-negotiation marketplace. Adoption skews toward already digitally comfortable farmers.
Private aggregators	Ninjacart, DeHaat, WayCool & similar	Re-centralize produce through their own collection agents — a new corporate middleman. Often require bulk volumes small farmers can't meet.

A pattern across all three: farmers consistently report needing immediate cash, distrusting delayed or unfamiliar digital payment, and preferring the certainty of a physically present buyer — even a middleman — over an unseen one. Any workable solution has to design around this, not assume it away.

3. Proposed solution: AgriConnect TN

AgriConnect TN is a direct listing and matching platform that lets a farmer (or someone posting on their behalf) publish an upcoming harvest before it is even picked, so buyers and transporters can plan around it instead of reacting to it after the fact. The system is deliberately built around three constraints existing systems ignore: uneven digital literacy, unreliable price reporting, and the farmer's need for real bargaining leverage.

3.1 Hybrid access — no one is forced onto a smartphone

A smartphone-comfortable farmer can register and post directly. A less digitally confident farmer's listing is created by a facilitator — a Village Level Entrepreneur at a Common Service Centre, or an FPO staff member — acting on their behalf, over a phone call or in-person visit. Both paths write to the same backend, so the pool of listings and price data stays unified regardless of who entered it.

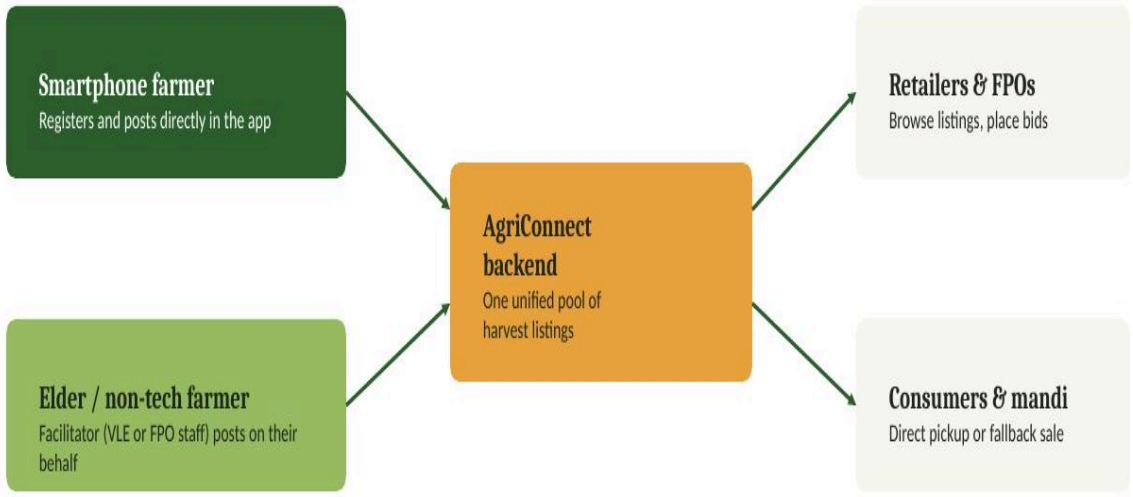


Figure 1. Two ways into the same system.

3.2 Pre-harvest listings, not post-harvest reactions

A listing moves through a simple lifecycle: Draft, Posted, Interest received, Confirmed, and Fulfilled. Fields are kept minimal — photos, an expected harvest window (a date range, not a fixed day, since ripening shifts), and an approximate quantity — and nothing about updating the listing is mandatory. When a listing is marked Fulfilled, the platform records the actual sale price and a mutual rating, which becomes the raw material for the platform's own price-transparency data over time.



Figure 2. The harvest listing lifecycle.

3.3 One listing, four buyer paths

A retailer, an FPO's bulk buyer, a group of consumers, and a mandi trader all want different things from the same harvest. Rather than forcing every sale through one channel, a single listing can be matched to whichever of four paths produces a real buyer within the farmer's harvest window:

- Retailer — a local shop buys for regular resale, e.g. a vegetable shop in Madurai placing a standing weekly order.
- FPO / bulk buyer — several farmers' listings are aggregated by their FPO and sold in bulk to a hotel, exporter, or processing unit.
- Consumer group — a residential community pools an order directly from the listing and arranges one shared pickup, with no reseller margin at all.
- Mandi (fallback) — if no other buyer appears in time, the produce still goes to the mandi as before, but the farmer now walks in already knowing a real reference price.

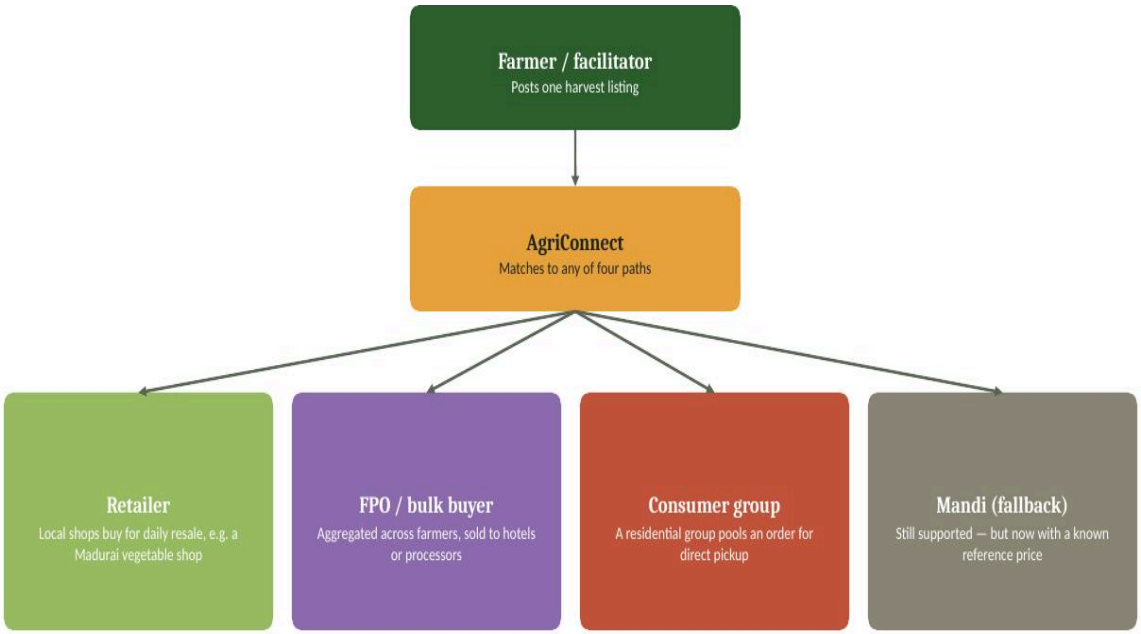


Figure 3. One listing branching into four buyer types.

3.4 Price transparency from three sources, not one

Official mandi price reporting is known to be inconsistent, and asking anyone to photograph a commission agent's price board risks real social friction, since price opacity is part of how that business model works. AgriConnect instead triangulates three sources and labels each by origin and recency rather than blending them into one unverifiable number: the government's own Agmarknet open dataset (legal, but often a few days stale), the platform's own fulfilled-listing sale prices (fresh and hyper-local, but a smaller sample), and optional self-reported personal experience from a farmer or facilitator's own recent transaction. Large mismatches between sources are flagged rather than silently resolved.

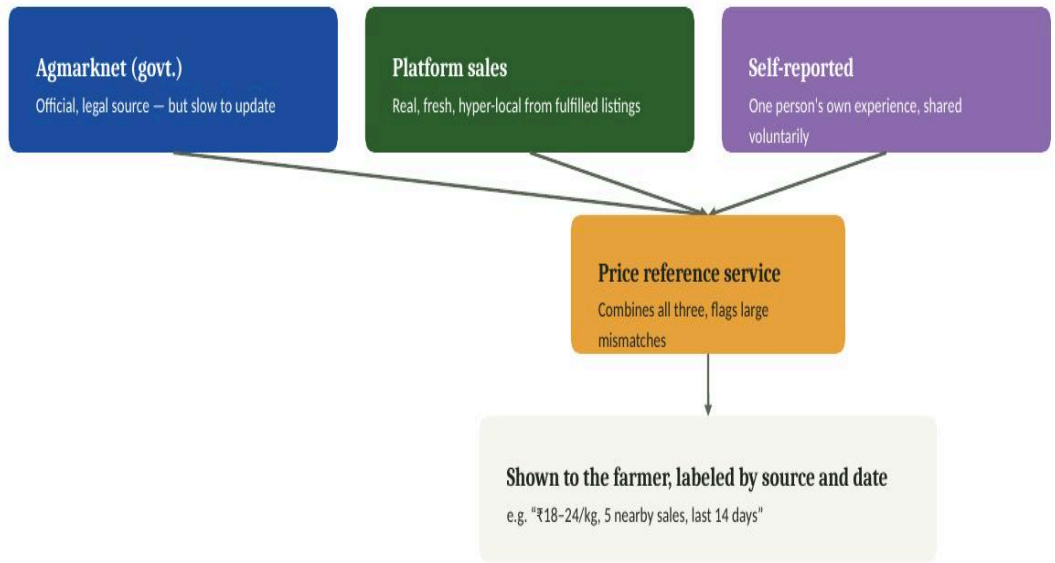


Figure 4. Price data triangulated from three labeled sources.

3.5 Keeping the design farmer-first

- The suggested price range is shown to the farmer before any buyer sees it, so the farmer sets the anchor rather than reacting to one.
- Buyers must state an offer price upfront within the platform — no private, off-platform negotiation channel.
- The farmer or facilitator always has final confirmation authority; the system never auto-accepts the lowest bid.
- Listing produce is free for the farmer; any future platform fee would apply to the buyer side.
- Ratings apply to buyers and transporters, not just to farmers — letting a farmer flag an unreliable buyer publicly.

4. Technical approach

The backend is a Spring Boot application, deliberately scoped to run entirely on free-tier infrastructure available through Azure for Students and the GitHub Student Developer Pack, so the project incurs no ongoing cost during development.

Backend	Spring Boot 3, Spring Security (JWT), Spring Data JPA
Database	PostgreSQL — Azure free tier or Neon.tech
Cache	Redis — Upstash free tier
Hosting	Azure App Service, F1 free tier
CI/CD	GitHub Actions, automatic deploy on push
Docs & monitoring	OpenAPI/Swagger, Azure Application Insights

Core entities: User (role-based — Farmer, Facilitator, Buyer, Transporter, Admin), HarvestListing, Interest, PickupSchedule, Outcome, and PriceReference.

5. Build roadmap

- Phase 1 — Set up accounts and tools (Azure for Students, IntelliJ, GitHub repository).
- Phase 2 — Scaffold the Spring Boot project with domain-organized packages.
- Phase 3 — Build the Auth and Farmer/Buyer module with JWT-based role access.
- Phase 4 — Add harvest listing creation and search.
- Phase 5 — Connect a free-tier cloud PostgreSQL database.
- Phase 6 — Add orders, FPO aggregation, and the price reference service.

- Phase 7 — Set up CI/CD and deploy to Azure App Service.
- Phase 8 — Polish for demo readiness: API docs, validation, README.

6. Why this matters

Every existing system reviewed here — e-NAM, Uzhavan, and private aggregators — solves part of this problem while leaving a real gap: stale or inaccessible price data, low reach among the least digitally confident farmers, or a new centralized middleman replacing the old one. AgriConnect TN is designed specifically around those documented gaps, so that more of the value in every harvest stays with the person who grew it — without requiring every farmer to become a smartphone power user to benefit.